

SYDNEY TECHNICAL HIGH SCHOOL

HSC ASSESSMENT

TASK 1

MATHEMATICS

December 2003

Time allowed: 70 minutes

Weighting 10%

Name: _____ Class: _____

Instructions

- Start each question on a new page
- Marks indicated are approximate only
- Answer all questions – marks may be deducted for poorly arranged work

leave out circled questions

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total

Question 1 (8 marks)

- a) For the series $-12 + 13 + 38 + \dots$
- Find the value of the 12th term (2)
 - Find the sum of the first 12 terms (2)
 - Which term has the value 963? (2)
- b) Write down the coordinates of the focus for the parabola $x^2 = 12 - 12y$ (2)

Question 2 (8 marks)

- a) The roots of $3x^2 - 5x - 4 = 0$ are α and β (6)

Find the value of,

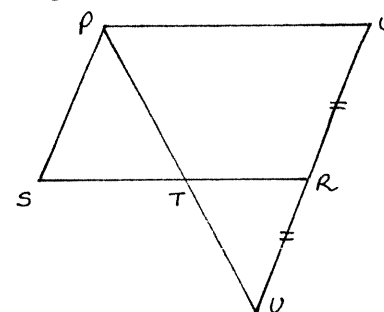
- $\alpha\beta$
- $\alpha + \beta$
- $\frac{1}{\alpha} + \frac{1}{\beta}$
- $\alpha^3\beta + \alpha\beta^3$

- b) The first two terms of a geometric series are 18 and 12 respectively. (2)

Find the limiting sum of the series.

Question 3 (8 marks)

- a) In the diagram, PQRS is a parallelogram. Q R is produced to U so that Q R = R U (4)



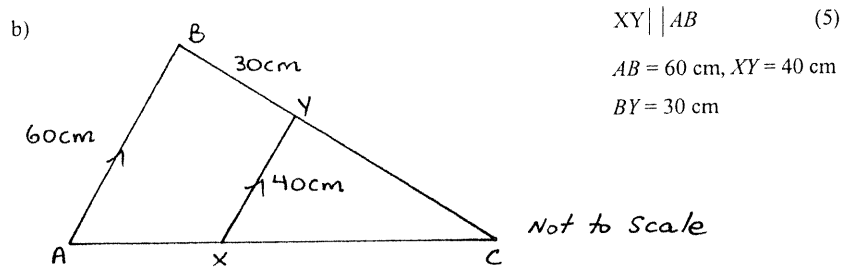
Not to Scale

- Giving clear reasons, show that the triangles PST and URT are congruent.
- Hence or otherwise, show that T is the midpoint of SR

- b) Write $0.\dot{1}0\dot{2}$ as a geometric series, and state the value of a and r . Hence, write $0.\dot{1}0\dot{2}$ as a (4)
fraction in its simplest form.

Question 4 (8 marks)

a) Evaluate $\sum_{n=4}^{20} 2n - 5$ (3)



- i Prove $\triangle ABC \sim \triangle XYC$
- ii Calculate the length CY

Question 5 (8 marks)

- a) Solve for K if the zeros of the parabola $y = 2x^2 - 7x + K$ are not real. (3)
- b) If $x + 2, 2x - 4, 4x + 1$ are the first three terms in an Arithmetic sequence, find x . (2)
- c) Find the values of P, Q and R if $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$ (3)

Question 6 (8 marks)

- a) A parabola has equation $x^2 = -12y$ (6)
- i Find the coordinates of the vertex of the parabola
- ii Write down the coordinates of the focus of the parabola
- iii Find the equation of the tangent to the parabola at the point where $x = 6$.
- iv Find the coordinates of M , the point where the tangent cuts the y axis
- b) Write down the equation of the parabola whose focus is $(1, 3)$ and directrix is $x = 5$. (2)

Question 7 (8 marks)

- a) Consider the points $A(-2, 1)$, $B(4, 1)$ and $P(x, y)$ (6)
- i Find expressions for the gradients of the two intervals PA and PB
- ii Show that the locus of P is $x^2 - 2x + y^2 - 2y = 7$ if $\angle APB = 90^\circ$
- iii Show that the locus represents a circle and give its centre and radius.

- b) Find the sum of the first ten terms of the series $1 - \sqrt{2} + 2 - \dots$ (2)

Express your answer in simplest exact form with a rational denominator.

Question 8 (8 marks)

- a) In an arithmetic series the eight term is 32 and the sum of the first ten terms is 400. (3)
- Find,
- i the value of the common difference
- ii the value of the first term
- b) At the beginning of each year Xena the Warrior princess invests \$1200 in a superannuation fund, on which she is paid 8.5% pa interest, compounded annually. (5)
- Find,
- i the amount of interest earned in the first year
- ii the total of her investments at the end of 25 years.

Teacher's Name:

Student's Name/N^o:2 unit HSC Ass one - 2003 - AnswersQuestion 1

a) $a = -12$ $d = 25$

i. $T_{12} = -12 + 11 \times 25$ — ① formula
 $= 263$ — ① Ans

ii. $S_n = \frac{n}{2}(a + t)$ ① formula
 $= \frac{12}{2}(-12 + 263)$ ① Ans
 $= 1506$

iii. $963 = a + (n-1)d$

$963 = -12 + (n-1) \times 25$ — ①

$975 = 25n - 25$

$1000 = 25n$

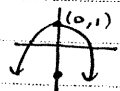
$n = 40$ — ①

 $\therefore 40^{\text{th}}$ term has value 963

b) $x^2 = -12y + 12$

$x^2 = -12(y-1)$ — ①

$V = (0, 1)$ $a = 3$



$S = (0, -2)$ — ①

Question 2

a) $a = 3$ $b = -5$ $c = -4$

i. $\alpha\beta = c/a = -4/3$ ①

ii. $\alpha + \beta = -b/a = 5/3$ ①

iii. $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$ — ①

$= 5/3 \div -4/3$

$= -5/4$ — ①

iv. $\alpha^3\beta + \alpha\beta^3 = \alpha\beta(\alpha^2 + \beta^2)$

$= \alpha\beta(\alpha + \beta)^2 - 2\alpha\beta$ — ①

$= -4 \left[\left(\frac{5}{3}\right)^2 - 2 \times -\frac{4}{3} \right]$ — ①

$= -7\frac{1}{27}$

Q2 (b) $18 + 12 + \dots$

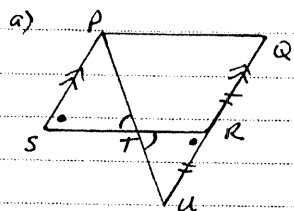
GP $a = 18$ $r = 12/18$

$= 2/3$ — ①

$S_{\infty} = \frac{a}{1-r}$

$= \frac{18}{1-2/3}$

$= 54$ — ①

Question 3i. $PS \parallel QR$ opp sides of Parallelogram equal — ① $PS = QR$ opp sides of Parallelogram equal.In ΔPST and ΔQRT

$PS = QR$ (= QR)

$\angle PST = \angle QRT$ (alt. $\angle$'s $PS \parallel QR$) — ①

$\angle PTS = \angle QTR$ (vert opp) — ①

$\therefore \Delta PST \equiv \Delta QRT$ (AAS)

ii. $ST = TR$ corresp. sides of $\equiv \Delta$'s $\therefore T$ is the Midpoint of SR — ①

b) $0.102 = \frac{102}{1000} + \frac{102}{1000000} + \dots$

$\therefore a = \frac{102}{1000}$ $r = \frac{1}{1000}$ ①

$S_{\infty} = \frac{a}{1-r}$ ①

$= \frac{102}{1000} \div \frac{999}{1000}$

$= \frac{102}{999}$

$= 34/225$ — ①

Teacher's Name:

Student's Name/N^o:Question 4

a) $\sum_{n=4}^{20} 2n - 5$

$n = 4$ $n = 5$ $n = 6 \dots n = 20$

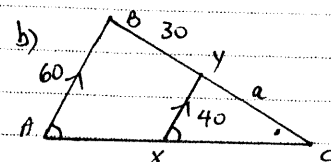
$3 + 5 + 7 + \dots + 35$ ①

AP $a = 3$ $d = 2$ $n = 20 - 4 + 1$
 $= 17$ ①

$S_n = \frac{n}{2}(a + t)$

$= \frac{17}{2}(3 + 35)$

$= 323$ ①

i. In $\Delta ABC \sim \Delta XYC$ $\angle C$ is common ①

$\angle BAC = \angle YXC$ (Corresp $\angle$'s $AB \parallel XY$) ①

$\therefore \Delta ABC \parallel \Delta XYC$ (all $\angle$'s =) ①

ii. $\frac{60}{40} = \frac{30+a}{a}$ ratio of ①
sides in $\parallel \Delta$'s

$6a = 120 + 4a$

$2a = 120$

$a = 60$

$\therefore CY = 60$ ①

Question 5

a) Not real $\Delta < 0$ ① $a = 2$
 $b^2 - 4ac < 0$ $b = -7$

$49 - 4 \times 2 \times K < 0$ ① $c = K$

$49 - 8K < 0$

$-8K < -49$

$8K > 49$

$K > 49/8$ ①

b) $x + 2, 2x - 4, 4x + 1, \dots$

AP $\rightarrow T_2 - T_1 = T_3 - T_2$ ①

$2x - 4 - (x + 2) = (4x + 1) - (2x - 4)$

$2x - 4 - x - 2 = 4x + 1 - 2x + 4$

$x - 6 = 2x + 5$

$x = -11$ ①

c) $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$

RHS $= P(x^2 + 2x + 1) + Qx + Q + R$

equating coefficients.

$* 3 = P$ ①

$* 5 = 2P + Q$

$5 = 6 + Q$ $Q = -1$ ①

$* -1 = P + Q + R$

$-1 = 3 - 1 + R$

$-1 = 2 + R$

$R = -3$ ①

$\therefore P = 3, Q = -1, R = -3$

Teacher's Name:

Student's Name/N^o:Question 6

a) $x^2 = -12y$

I. $V = (0, 0)$ ①

II. $S = (0, -3)$ ①

III. $y = \frac{x^2}{-12}$

$y' = \frac{2x}{-12}$ at $x=6$ $y=-3$ ①

$m_T = -1$

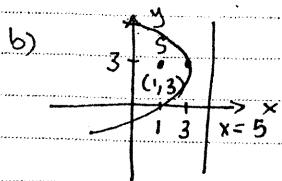
∴ equation

$y + 3 = -1(x - 6)$ — ①

$y + 3 = -x + 6$

$x + y - 3 = 0$

IV. $x = 0$ $y = 3$ ①

Vertex $(3, 3)$

$a = 2$

$(y - y_1)^2 = 4a(x - x_1)$

$(y - 3)^2 = -8(x - 3)$

① ↑ ① Neg.

Question 7

a) $m_{PA} = \frac{y-1}{x+2}$ $m_{PB} = \frac{y-1}{x-4}$ ① ①

II. $\frac{y-1}{x+2} \times \frac{y-1}{x-4} = -1$ ①

$\frac{y^2 - 2y + 1}{x^2 - 2x - 8} = -1$

$y^2 - 2y + 1 = -x^2 + 2x + 8$

$x^2 - 2x + y^2 - 2y = 7$ ①

III. $x^2 - 2x + 1 + y^2 - 2y + 1 = 7 + 2$

$(x-1)^2 + (y-1)^2 = 9$

$C = (1, 1)$ $r = 3$ ① ①

b) $1 - \sqrt{2} + 2 - \dots$

G.P. $a = 1$ $r = -\sqrt{2}$

$n = 10$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{1((- \sqrt{2})^{10} - 1)}{-\sqrt{2} - 1}$

$= \frac{1(32 - 1)}{-\sqrt{2} - 1}$

$= \frac{31}{-\sqrt{2} - 1}$

$= \frac{-31}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$

$= \frac{-31\sqrt{2} + 31}{1}$

$= -31\sqrt{2} + 31$

Teacher's Name:

Student's Name/N^o:Question 8

a)

$T_8 = a + 7d = 32$

$S_{10} = \frac{10}{2}(2a + 9d) = 400$

$2a + 9d = 80$ ①

$2a + 14d = 64$

$\therefore 5d = 16$

$d = \frac{-16}{5}$ ①

$a + 7x^{-16/5} = 32$

$a = 54\frac{2}{5}$ ①

b)

I. $A = 1200(1.085)^1$

$= 1302$ ①

$\therefore \text{Interest} = \102 ①

II. $A_1 = 1200(1.085)^{25}$

$A_2 = 1200(1.085)^{24}$

$A_3 = 1200(1.085)^{23}$

$\vdots$

$A_{25} = 1200(1.085)^1$ ①

$\text{Total} = 1200(1.085 + 1.085^2 + \dots + 1.085^{25})$

$= 1200 \times \frac{a(r^n - 1)}{r - 1}$

$a = 1.085$

$r = 1.085$

$n = 25$

① — setting up G.P.

$= 1200 \times \frac{1.085(1.085^{25} - 1)}{0.085}$

$= \$10\,2425.47$ — ①